

Prim's Algorithm Using Min Heap

Aim

To design and implement Prim's Algorithm to find the Minimum Cost Spanning Tree of a connected undirected graph using Min Heap.

Theory

Prim's Algorithm is a Greedy Algorithm used to find the Minimum Spanning Tree (MST) of a connected weighted graph.

The algorithm starts from a source vertex and repeatedly selects the minimum weight edge connecting visited and unvisited vertices.

In this implementation, Min Heap is used to efficiently select the minimum edge.

Algorithm

1. Convert input graph into Adjacency Matrix.
2. Take result matrix (res) of size $V \times V$ and initialize with zeros.
3. Take visited array (vis) of size V and initialize with zeros.
4. Take Min Heap and initialize heap size as zero.
5. Read the source vertex (sv). Example: $sv = 0$.
6. Change $vis[sv] = 1$.
7. Go to adjacency matrix row of source vertex.
8. Consider non-zero values and insert corresponding edges into Min Heap.
9. Min Heap contains four parts: x, y, w where:
 - x = source vertex
 - y = destination vertex
 - w = edge weight
10. Heap is maintained based on minimum weight (w).
11. Initialize count = 1.
12. Repeat until count becomes V :
 - a) Delete minimum edge from heap and store in temp.

- b) If destination vertex y is already visited, move to next iteration.
 - c) Consider vertex y from temp.
 - d) Go to adjacency matrix row of y .
 - e) Consider non-zero and unvisited adjacent vertices.
 - f) Insert corresponding edges into Min Heap.
 - g) Change $vis[y] = 1$.
 - h) Increment count ($count++$).
 - i) Store deleted temp edge into result matrix.
13. Display result matrix.
14. Calculate total minimum weight.
15. Stop.

Time Complexity

Using Min Heap, Prim's Algorithm works efficiently.

Time Complexity = $O(E \log V)$

Where:

E = Number of edges

V = Number of vertices

Result

Thus, the Minimum Cost Spanning Tree of the given graph is obtained successfully using Prim's Algorithm with Min Heap.